ChatGPT

ASO Multi-Domain Research Brief Validation and Execution Plan

Executive Summary

The brief is a **multi-domain ASO research specification**, not a simple keyword-audit prompt. It combines App Store Optimization, creative conversion analysis, regional localization strategy, competitor intelligence, review mining, and open-source tooling due diligence. In other words, it is broad enough to justify a Deep Research or Pro Extended-style run, because several parts of the brief require cross-checking official Apple and Google documentation, live store-page observation, regional segmentation, and code/repo validation rather than a single-source answer. filecite turn0file0 [1]

My readiness verdict is **conditionally ready**: the brief is strong on scope, methods, and desired deliverables, but it is still missing a few constraints that materially affect research quality. The most important gaps are the **target app list**, the **category or categories to benchmark**, the **exact country-language matrix**, the **success KPI for ROI scoring**, and the **exact GitHub repo URLs** for the audit portion. If those are not supplied, the best conservative fallback is the one already implied in your instruction: identify the top competitors by category and proceed with a transparent “assumed competitor universe” section. filecite turn0file0 [2]

Two assumptions in the current brief should be revised before the main run. First, **hard-coded locale counts** such as “Apple has 40 locales” and “Google Play has 39” should not remain fixed claims. Apple’s current App Store localization documentation lists a much larger set of supported localizations, and Google Play Console currently lists a long set of self-managed translation languages extending far beyond 39. The safer phrasing is: **“verify the current localization set at execution time.”** [3]

Second, the brief should distinguish **store-search indexing**, **web search indexing**, and **personalized listing delivery**. Apple’s official metadata docs define the app name, subtitle, dedicated keywords field, promotional text, and description separately, and explicitly say the description is used for **web engine search results**. Google’s official docs show that Play supports keyword-targeted custom store listings, country targeting, and search-keyword variation bundles, which means result pages can differ by country, query, and audience segment. A rigorous research run therefore cannot treat “indexing” as a single monolithic thing. [4]

What the Brief Is and Whether It Is Ready

At a high level, the brief is best classified as **ASO plus adjacent market intelligence**. It spans both major stores, asks for Apple-versus-Google indexing verification, requires regional behavior analysis, and adds GitHub repo audits and review sentiment analysis. That makes it broader than classic ASO, and closer to a hybrid of **store-growth research, localization research, and tooling-risk assessment**. Apple’s and Google’s own documentation also supports that broader framing, because store performance on both platforms depends not only on text metadata but also on screenshots, video, localizations, search presentation, and testing frameworks. [5]

The brief is already strong in several ways. It asks for recent sources, prioritizes official documentation, requires explicit deliverables, requests regional segmentation, and includes methods that are appropriate for this type of work: manual page review, review analysis, code/repo review, and web-based evidence gathering. Those are the right ingredients for a serious ASO research program, and they align well with current platform realities such as Apple's product page optimization and Google's custom store listings and language-localization workflows. fileciteturn0file0 [6]

Where the brief is **not fully ready** is in its missing decision rules. Category research is much more useful when it is done within a clearly defined niche rather than across the entire store universe; recent app-store research likewise finds that category-focused analysis yields more actionable insight than undifferentiated app-store-wide scanning. So, before the main run, the research brief should lock one of these two routes: either a supplied list of target apps, or a clearly defined category/problem space from which competitors will be chosen. [7]

A concise readiness matrix is below.

Dimension	Status	Assessment
Research type	Ready	Multi-domain ASO research with localization, competitor analysis, review mining, and OSS/tooling audit. filecitetturn0file0
Timeframe	Ready	"Prefer 2024–2026 sources" is a strong default, with older primary docs allowed only where still authoritative. filecitetturn0file0 [8]
Platforms	Ready	Apple App Store and Google Play are explicit, and platform differences are research-relevant. filecitetturn0file0 [9]
Deliverables	Ready	Long-form report, 30+ citations, competitor metadata table, ROI prioritization, regional behaviors, GitHub audits. filecitetturn0file0
Target app list	Missing	The brief assumes it will be provided, but the main run needs either the list or a locked competitor-selection rule. filecitetturn0file0
Geographic scope	Partial	"Global" is a fair default, but the primary countries and language variants still need to be fixed operationally. filecitetturn0file0 [10]
KPI for ROI scoring	Missing	Without a primary KPI, ROI ranking can drift between visibility, conversion, retention, and monetization.
GitHub repo identifiers	Missing	Repo names alone are ambiguous; the audit requires exact URLs. filecitetturn0file0 [11]

Missing Constraints and Risks

The single biggest missing constraint is the **target-app universe**. If the user's apps are not named, the research must infer the relevant competitor set from category, use case, and install band. That is feasible, but it changes the nature of the deliverable: the final report would need an explicit "competitor selection logic" section so that the benchmark set is reproducible and contestable. That is especially important because current app markets are dynamic and apps can rebrand, repurpose, or materially shift positioning over time. [12]

The second major gap is the **country-language matrix**. "Global" is acceptable as a default research assumption, but the brief also points toward specific markets. Both Apple and Google make it clear that localization is not just a matter of translation. Apple ties localized metadata to searchability by localized keywords, while Google supports both language localization and separate country-targeted custom store listings. That means the final brief should explicitly declare whether the unit of analysis is country, language, country-language pair, or some combination of those. [13]

A third risk is **conflating indexing with presentation**. On Apple, the app name, subtitle, keywords field, promotional text, and description are different fields with different constraints; Apple also states that the description is used for web engine search results. On Google Play, the short description is limited to 80 characters, and Google explicitly warns that unnecessary keywords in that field "will not impact ranking," while custom store listings can target search keywords and expose spelling-correction and translation variations. In practice, that means the final research should split platform verification into at least three layers: **on-store search relevance signals**, **page-conversion assets**, and **web/deep-link indexing**. [14]

A fourth constraint is **data asymmetry across platforms**. Public Google Play-facing tooling and wrappers routinely surface install bands and rich app-page details, while iOS-facing wrappers generally surface ratings, reviews, versions, screenshots, and metadata but not public install counts. That means the brief should not require symmetrical "install delta" fields for Apple and Google. For iOS, competitor scale should be proxied using review volume, rating density, chart presence, app age, update cadence, localization breadth, and asset sophistication. This is an inference from currently accessible public wrapper outputs and store-page structures, and it should be stated as such in the final brief. [15]

A fifth constraint is **tooling fragility and legal/operational access**. The popular `google-play-scraper` repository explicitly says the maintainer no longer actively maintains it and warns that the parser may break when Google Play's layout changes, even though community contributions continue and the commit history shows updates into late 2025. Apple's official iTunes Search API documentation is also archived and last updated in 2017, so any repo audit needs to separate "wrapper still works" from "upstream documentation is current." The brief should therefore require every tooling recommendation to include maintenance recency, stars/forks, issue backlog, and parser-breakage risk. [16]

A compact risk table is below.

Missing item or risk	Why it matters	Conservative default
Hard-coded locale counts	Apple and Play currently expose broader localization support than the fixed counts in the brief imply. [3]	Say "verify current supported localizations at execution time."
On-store vs web indexing ambiguity	Apple description is documented for web search results; Google has a separate Google Search app-indexing workflow. [17]	Separate "store search," "web search," and "deep-link indexing" in the report.

Missing item or risk	Why it matters	Conservative default
Query-personalized or keyword-targeted listings	Google custom store listings can target search keywords, countries, URLs, and user state. [18]	Capture search results by query, country, and language; note variant risk.
Asset/testing interaction	Apple PPO treatments can surface in search and run up to 90 days with up to three treatments. [19]	Treat creative testing as part of ASO, not as a separate appendix.
Screenshot update friction on Apple	After app approval, screenshot updates require a new version. [20]	Flag screenshot changes as higher-friction than promo-text changes.
Repo-name ambiguity	Names like `itunes-search-api`, `asoreview`, and `openaso` are not enough for reliable audit.	Require exact repo URLs before final repo scoring.

Deliverables and Tables to Require

The final research output should absolutely begin with an **executive summary**, because the brief is too large for the reader to derive the priority actions themselves. The summary should answer four questions up front: what was verified, what changed relative to the original assumptions, which competitors matter most, and what the top ROI actions are. That structure is justified by the volume of platform-specific mechanics in Apple and Google documentation and by the need to reduce a multi-source audit into decisions. [21]

The first required table should be a **competitor comparison table**. Because category-focused analysis is more actionable than broad-store scanning, this table should make the competitor-selection logic explicit, rather than silently mixing category leaders, direct substitutes, and small entrants. [7]

Competitor tier	Inclusion logic	What to extract	Why it matters
Direct competitors	Same use case and same category/ problem-to-be-done	Title, subtitle/short description, screenshots, video, ratings, reviews, update cadence, pricing, localization coverage	Core benchmark set
Aspirational leaders	Same category, much larger scale	Win patterns in metadata, creative strategy, localization breadth, social proof	Ceiling-setting examples
Low-end or emerging apps	Same category, lower traction	Gaps they leave, sloppy metadata, weak creative execution	“Winnability” comparisons
Adjacent substitutes	Different category but same user need	Alternate positioning, alternative keywords, CRO ideas	Discover hidden keyword and positioning threats

Competitor tier	Inclusion logic	What to extract	Why it matters
Regional variants	Same function, country-specific or language-specific emphasis	Regionally tailored screenshots, message framing, feature emphasis	Localization playbooks

The second required table should be a **deliverables and sequencing table**. Since actual timing depends on access, app count, and market scope, the table should describe execution order rather than promise elapsed durations.

Deliverable	Minimum contents	Primary evidence base	Sequence
Executive summary	What changed, what matters, top actions	Full synthesis	First in report
Store mechanics verification	Apple vs Google field constraints, indexing distinctions, testing mechanics	Official Apple/Google docs	Early
Competitor metadata table	Structured benchmark set with country/language notes	Store pages, scrapers, manual capture	Early-mid
Review sentiment analysis	Theme clusters, sentiment mix, repeat pain points, localization complaints	Store reviews, review-mining methods	Mid
GitHub/tooling audit	Repo recency, stars, maintenance risk, language/country support	GitHub + official docs	Mid
Regional behavior analysis	Country/language differences, listing patterns, message fit	Localized pages + official localization docs	Mid-late
ROI prioritization	Ranked opportunities with effort, confidence, expected impact	Synthesized evidence	Late
Evidence appendix	30+ citations with source quality notes	All sources	Final appendix

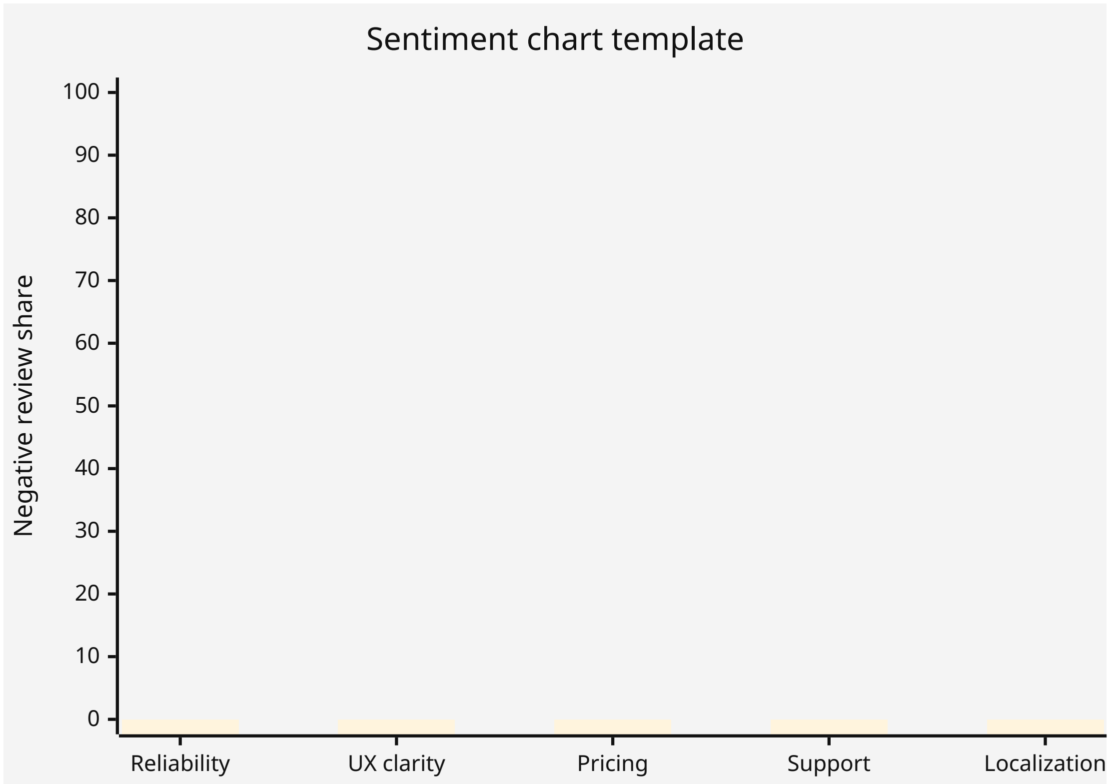
The third required table should be a **source list table** that tells the reader exactly what source hierarchy the research followed.

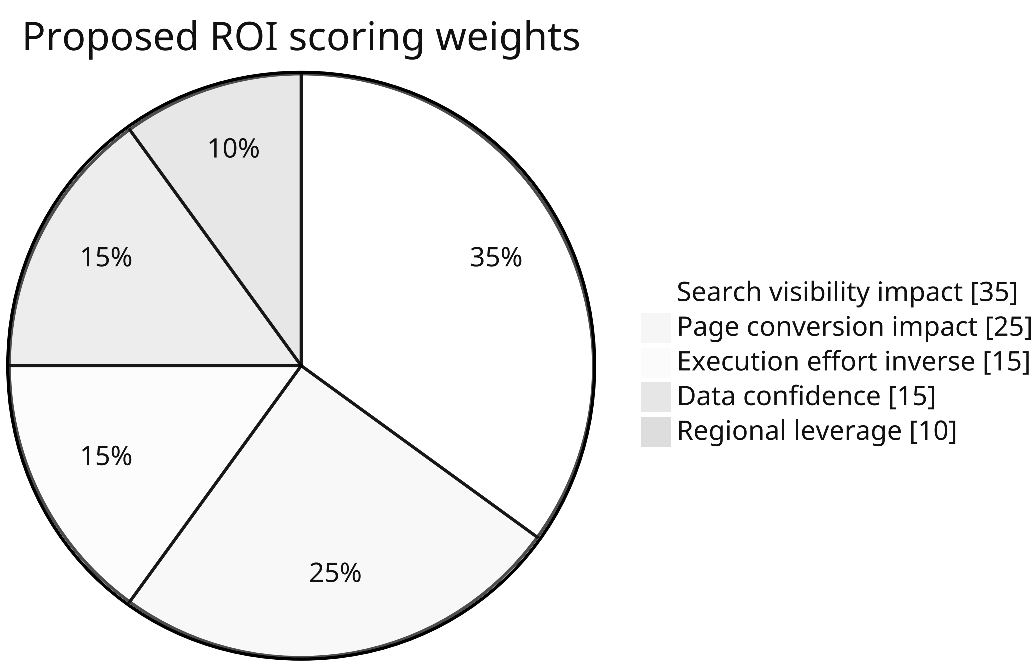
Priority band	Source family	Why it should be prioritized	Example verified sources
Highest	Official Apple docs	Canonical for App Store fields, localizations, screenshots, PPO, review constraints	App information; platform version information; localize app information; upload screenshots; PPO. [22]

Priority band	Source family	Why it should be prioritized	Example verified sources
Highest	Official Google docs	Canonical for Play localization, preview assets, custom listings, and app indexing	Translate/localize app; preview assets; custom store listings; app indexing. [23]
High	GitHub repos	Needed for scraper and tooling feasibility checks	`google-play-scraper`, `app-store-scraper`, and exact repo URLs for others. [24]
High	Primary technical docs/ APIs	Useful for wrapper drift and API feasibility	iTunes Search API archive. [8]
Medium	Recent academic papers	Strong support for review mining, category-scoped analysis, dynamic-market caution	Category study, app metamorphosis, review prioritization, description-behavior mismatch. [25]
Medium	Community and vendor blogs	Useful for edge cases and field observations, but must be triangulated	Apple Developer Forums, Reddit, AppTweak, Phiture, SplitMetrics, App Radar, Mobile Action.
Optional	Paid market intelligence	Helpful for revenue and geo-sizing, if access exists	Sensor Tower, data.ai, Appfigures.

Proposed visualization requirements

The final report should include a **sentiment chart** and an **ROI chart**, but because those depend on execution data, the correct thing to define now is the output specification rather than fake data.


Sentiment chart template
Negative review share
100
90
80
70
60
50
40
30
20
10
0
Reliability
UX clarity
Pricing
Support
Localization


Proposed ROI scoring weights
10%
15%
35%
15%
25%
Search visibility impact [35]
Page conversion impact [25]
Execution effort inverse [15]
Data confidence [15]
Regional leverage [10]

The ROI weighting above is a **proposed default scoring model**, not an empirical result. It should be adjusted once the primary KPI is known.

Recommended Research Method

The research flow should begin with **mechanics verification**, not with competitor screenshots. Apple and Google both have enough platform-specific rules that a competitor page can be misread if the field constraints are not understood first. Apple's official docs now verify that the app name is capped at 30 characters, the subtitle at 30 characters, promotional text at 170 characters, the description at 4000 characters, and keywords at up to 100 bytes; Apple also documents that apps can be searched using localized keywords. Google's docs verify an 80-character short description, warn that stuffing unnecessary keywords into that field does not improve ranking, and document search-keyword-targeted custom store listings. These changes are foundational to any later benchmark table. [26]

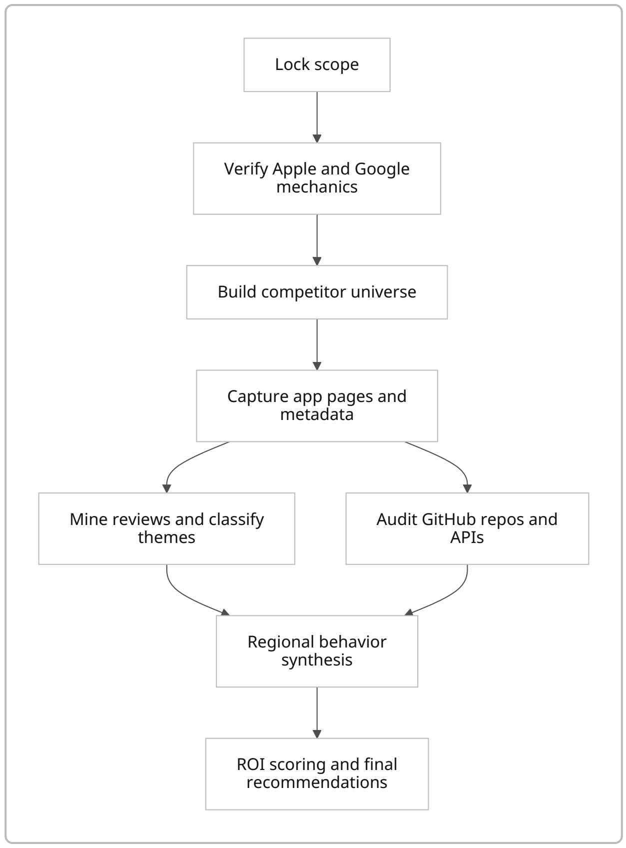
The second phase should build a **competitor universe by category, country, and query context**. That matters because Google custom store listings can vary by country, search keyword, URL parameter, and user state, and Apple product page optimization can place alternate assets in search and browse surfaces. So the research must capture competitors under standardized conditions such as fixed query strings, fixed countries, fixed device context, and fixed language settings wherever possible. This is partly a methodological inference from official platform docs, but it is a very strong one. [27]

The third phase should combine **manual store-page review with lightweight scraping and structured extraction**. Public scraping wrappers are useful, but they should be used carefully. `google-play-scraper` is still updated by community contributors but warns that layout changes can break it; `app-store-scraper` is viable for iOS page fields; and Apple's archived iTunes Search API docs remind us that wrapper maintenance and upstream-doc freshness are separate questions. The final report should therefore distinguish "evidence directly observed on live store pages" from "evidence extracted through wrappers." [28]

The fourth phase should run **review sentiment and topic analysis**. That is justified both practically and academically. Google Play review-prioritization research shows there is value in triaging developer-response-worthy reviews, and app-category studies show topic modeling and sentiment analysis can uncover the negative themes that actually affect competitiveness. This should not be limited to generic positive/negative sentiment; the report should isolate categories such as bugs, onboarding confusion, pricing friction, trust/privacy concerns, support delays, and localization complaints. [29]

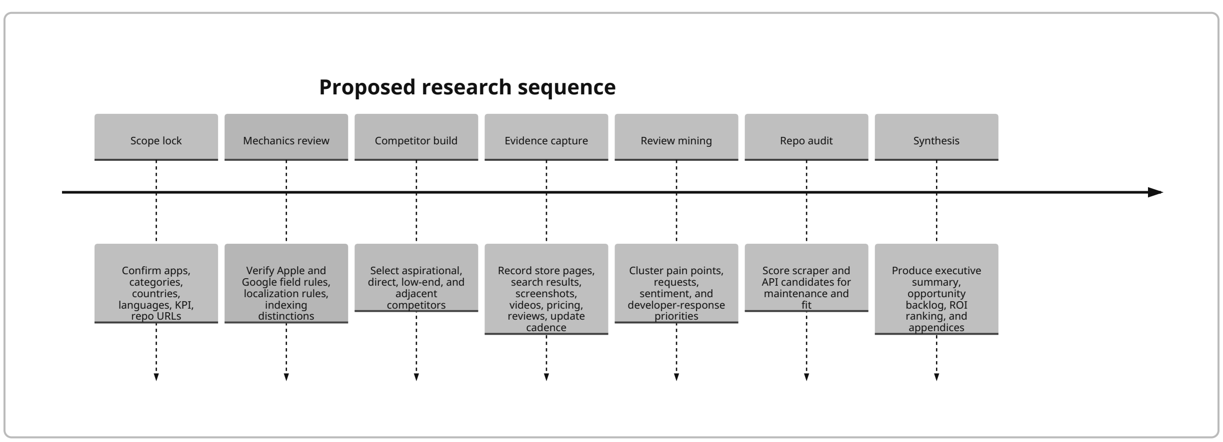
The fifth phase should conduct a **GitHub repo audit**, not merely a repository list. The audit criteria should include repository purpose, stars/forks, last commit recency, issue backlog, declared maintenance status, license, language/locale parameters, country support, output fields, rate limiting, and whether the repo's assumptions still match present-day store layouts. Since app descriptions can diverge from actual behavior, code/repo review is a useful complement to app-page analysis rather than a distraction from it. [30]

The sixth and final phase should synthesize everything into an **ROI-prioritized action backlog**. Apple-specific quick wins and Google-specific quick wins should not be lumped together. For example, Apple's promotional text can be updated without requiring an updated submission, while screenshot changes are more operationally constrained; Google's localization recommendations are explicitly based on category, install growth, conversion rate, and market potential, which makes them useful evidence for regional prioritization. [31]



Proposed Execution Sequence

The research sequence below is the right level of detail for the main brief. It defines the workstream order without inventing elapsed time estimates.



Operationally, I would recommend this **fallback rule** if the target list is still missing during the main run: select the top competitors by category using a reproducible blend of search visibility, ratings/reviews, localization breadth, update recency, and relevance to the same user job-to-be-done, then label the whole benchmark set as **"assumed competitor universe"** on the first page of the report. That keeps the research honest without blocking execution. [2]

Open Questions and Limitations

A few items should remain explicitly open rather than being guessed.

The first is the **exact target app list and category set**. The plan above is ready, but the benchmark quality will improve materially if specific apps and categories are locked before the main run.

The second is the **exact GitHub identities** for repositories named only generically in the brief, especially `itunes-search-api`, `asoreview`, and `openaso`. Two repositories were easy to verify and are suitable baseline candidates: `facundoolano/google-play-scraper` and `facundoolano/app-store-scraper`. [24]

The third is the **current public Google Play title and full-description character-limit help page**. This pass verified the short-description limit and the existence of title, short description, and full description in Play's localization workflow, but I did not confirm a current public official help page for every Play text-field limit during this validation pass. That gap should be closed at the start of the main run. [32]

The fourth is the **Apple Search Ads popularity/help-page reference** requested in the brief. The brief is right to include it as a source class, but I did not confirm a public official Apple help URL for that specific metric during this validation pass. It should remain in scope, but as a verification target rather than a fixed assumption.

Overall verdict: **the brief is understood, structurally strong, and ready for a serious deep-research run once the missing constraints above are made explicit.** The largest changes I would make before launching the full research are: remove hard-coded locale counts, split indexing into store/web/personalized-listing layers, define the competitor-selection rule, and require exact repo URLs for GitHub auditing. [33]

---

[1] [5] [6] [19] [21] [27] Product Page Optimization - App Store - Apple Developer
https://developer.apple.com/app-store/product-page-optimization/

[2] [7] [25] A Large-Scale Exploratory Study of Android Sports Apps in the Google Play Store
https://arxiv.org/abs/2310.07921?utm_source=chatgpt.com

[3] [33] App Store localizations - App information - Reference - App Store Connect - Help - Apple Developer
https://developer.apple.com/help/app-store-connect/reference/app-information/app-store-localizations

[4] [9] [14] [22] [26] App information - App information - Reference - App Store Connect - Help - Apple Developer
https://developer.apple.com/help/app-store-connect/reference/app-information/app-information

[8] iTunes Search API: Overview
https://developer.apple.com/library/archive/documentation/AudioVideo/Conceptual/iTuneSearchAPI/index.html

[10] [18] Create custom store listings to target specific user segments - Play Console Help
https://support.google.com/googleplay/android-developer/answer/9867158

[11] [15] [16] [24] [28] [30] GitHub - facundoolano/google-play-scraper: Node.js scraper to get data from Google Play · GitHub
https://github.com/facundoolano/google-play-scraper

[12] Detecting and Characterising Mobile App Metamorphosis in Google Play Store
https://arxiv.org/abs/2407.14565?utm_source=chatgpt.com

[13] Localize app information - Manage app information - App Store Connect - Help - Apple Developer
https://developer.apple.com/help/app-store-connect/manage-app-information/localize-app-information

[17] [31] Platform version information - App information - Reference - App Store Connect - Help - Apple Developer
https://developer.apple.com/help/app-store-connect/reference/app-information/platform-version-information

[20] Upload app previews and screenshots - Manage app information - App Store Connect - Help - Apple Developer
https://developer.apple.com/help/app-store-connect/manage-app-information/upload-app-previews-and-screenshots

[23] Translate and localize your app - Play Console Help
https://support.google.com/googleplay/android-developer/answer/9844778

[29] Prioritizing App Reviews for Developer Responses on Google Play
https://arxiv.org/abs/2502.01520?utm_source=chatgpt.com

[32] Add preview assets to showcase your app - Play Console Help
https://support.google.com/googleplay/android-developer/answer/9866151